REAL ESTATE INVESTMENT ADVISOR

Predicting Property Profitability & Future Value

BCA Project Report

Prepared as an educational machine-learning prototype

Abstract

This project demonstrates a machine-learning workflow for estimating property prices and exploring investment scenarios through a Streamlit dashboard. A Random Forest regression model predicts sample property price and a Random Forest classifier identifies properties at or above the sample dataset median. The dashboard also calculates a user-defined future-value scenario.

Problem Statement

Property buyers and students need a structured way to compare property attributes and understand how assumed growth and rental yield affect a scenario. This project demonstrates data preprocessing, machine learning, visual analytics and a web interface.

Objectives

Prepare a structured dataset; explore price patterns; build regression and classification models; evaluate model results; create a Streamlit interface; and track training runs with MLflow.

Technology Stack

Python, Pandas, NumPy, scikit-learn, Matplotlib, Streamlit, MLflow and CSV.

Dataset and Features

The included dataset has 1,200 synthetic records. Features include city, area, bedrooms, bathrooms, property age, location score, amenities and annual rent yield. The target is price in INR. The dataset is synthetic and is not verified market data.

Methodology

The workflow includes data inspection, EDA, an 80/20 train/test split, median imputation and scaling for numeric features, most-frequent imputation and one-hot encoding for city, Random Forest regression, Random Forest classification, and evaluation with MAE, RMSE, R-squared and accuracy.

Future Value and Profitability Scenario

Future Value = Predicted Current Value × (1 + Annual Growth Rate)^Number of Years. Projected gain equals future value minus the user-entered purchase price. ROI is projected gain divided by purchase price, multiplied by 100. Taxes, registration, maintenance, financing, vacancy and transaction costs are excluded.

Evaluation Metrics

MAE is the average absolute prediction error. RMSE penalizes larger errors more strongly. R-squared measures variance explained on held-out test data. Classification accuracy is the fraction of correct above-median-class predictions. Actual values are calculated when the app runs.

Results and Discussion

The app contains a property input panel, estimated price, future value scenario, illustrative annual rent, EDA visualizations and model evaluation. Metrics reflect only the synthetic educational dataset and do not prove real-market performance.

Limitations

Synthetic data is not representative of actual prices; location score is a simplified proxy; future value assumes constant user-selected growth; ownership costs and legal due diligence are not modeled; and classification is based on the sample median rather than investment suitability.

Future Scope

Use verified local listings or transaction data, add locality and geospatial attributes, include EMI and ownership costs, test gradient boosting, use time-aware validation for dated records, and add model explainability and monitoring.

Conclusion

The project demonstrates an end-to-end ML application involving data preparation, EDA, feature engineering, regression, classification, evaluation, MLflow tracking and an interactive Streamlit dashboard. It is a learning prototype and should be trained on verified real-world data before real investment use.

Sample project structure

File	Purpose
app.py	Streamlit dashboard
train_model.py	Training, evaluation, saved models and MLflow
data/housing_data.csv	1,200-row synthetic dataset
requirements.txt	Required Python packages
README.md	Setup and run instructions

Academic honesty and use note: the bundled dataset is synthetic, and model results are educational examples. Do not present these as verified real-estate market findings.